



Logarithms

Just as taking a square root is the opposite of squaring a number, the opposite of raising a base to a variable power is called taking a logarithm.

A **logarithm** is the exponent to which a base would have to be raised to result in a given value. Using logarithms, you would write $2^x = 10$ as $x = \log_2 10$.

Definition

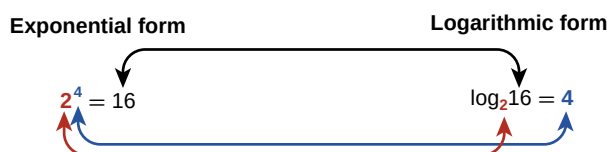
The logarithm of a with base b , $\log_b a$, where $b > 0$, $b \neq 1$, and $a > 0$, is defined as $\log_b a = x$ if and only if $b^x = a$.

► Think About It

NOTATION The expression $\log_b a$ is read as “log base b of a .”

Converting Between Logarithmic and Exponential Forms

The definition of a logarithm can be used to convert between exponential and logarithmic forms.



The table shows equivalent equations written in both exponential and logarithmic forms.

Exponential form	$5^2 = 25$	$4^{-3} = \frac{1}{64}$	$81^{\frac{1}{2}} = 9$	$10^0 = 1$
Logarithmic form	$\log_5 25 = 2$	$\log_4 \left(\frac{1}{64} \right) = -3$	$\log_{81} 9 = \frac{1}{2}$	$\log_{10} 1 = 0$

Logarithms with special bases have names.

Definitions

A **common logarithm** is a logarithm with base 10. Common logarithms, such as $\log_{10} x$, are usually written without the base, as $\log x$.

A **natural logarithm** is a logarithm with base e . Natural logarithms, such as $\log_e x$, are often written using the notation $\ln x$.

► **Remember** Like π , the number e is an irrational number.

$$e \approx 2.71828 \dots$$

EXAMPLE 1

Evaluate.

A $\log 0.001$

SOLUTION

Since the base of the log is not stated, the base is 10. Let x be the value of $\log 0.001$.

$$\log 0.001 = x$$

$$10^x = 0.001 \quad \text{Rewrite equation in exponential form.}$$

$$10^x = 10^{-3} \quad \text{Rewrite 0.001 as a power of 10.}$$

$$x = -3 \quad \text{Solve for } x.$$

B $\ln \sqrt{e}$

SOLUTION

Since the $\ln$ notation is used, the base is e . Let x be the value of $\ln \sqrt{e}$.

$$\ln \sqrt{e} = x$$

$$e^x = \sqrt{e} \quad \text{Rewrite equation in exponential form.}$$

$$e^x = e^{\frac{1}{2}} \quad \text{Rewrite } \sqrt{e} \text{ as a power of } e.$$

$$x = \frac{1}{2} \quad \text{Solve for } x. \quad \blacksquare$$

Using a Calculator to Find Common and Natural Logarithms

Calculators are programmed to calculate common and natural logarithms.

► **Think About It** Pietro Mengoli and Nicholas Mercator called the natural logarithm *logarithmus naturalis*.

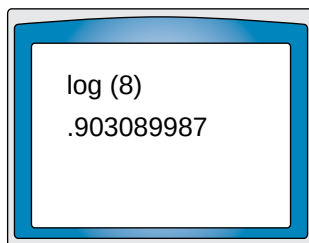
EXAMPLE 2

Use a calculator to evaluate the logarithm to four decimal places.

A $\log 8$

SOLUTION

On a calculator, press **LOG**, type **8**), and press **ENTER**.



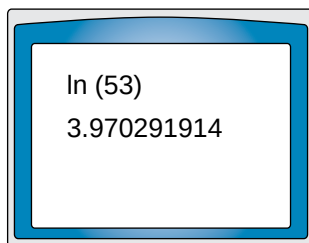
$$\log 8 \approx 0.9031$$

- **Think About It** On a scientific calculator, you often have to enter the value and then press the function key. For instance, **8** then **LOG**.

B $\ln 53$

SOLUTION

On a calculator, press **LN**, type **53**), and press **ENTER**.



$$\ln 53 \approx 3.9703 \quad \blacksquare$$

- **Think About It** The **LOG** key on a calculator is usually a common logarithm with base 10.
The **LN** key on a calculator is a natural logarithm with base e .

Using Logarithms in a Formula

EXAMPLE 3

The time t it takes for an investment with continuous growth rate r to double is given by the formula $t = \frac{\ln 2}{r}$. How many years would it take for an investment to double with a continuous rate of 3%?

SOLUTION

A rate of 3% means that $r = 0.03$. Substitute the value of r into the formula.

$$t = \frac{\ln 2}{0.03} \approx \frac{0.693}{0.03} \approx 23.1$$

It will take approximately 23 years for the investment to double. $\blacksquare$